A. Value Calculation under Present Value Ordinary Annuity:

Corporate

Example- 06

Manha Company needs to finance a machine that will cost Tk. 335,220. The company has two alternatives to finance the equipment as follows:

(i) To take lease from Ideal Leasing Company for 5 years with an annual lease payment of Tk. 120,000 to be paid at the end of each year.

(ii) To take loan from Baraka Bank at 15% interest repayable at the end of each year for 5 years. The equipment will be depreciated on straight line basis. The corporate tax rate is 40%. Which offer should be accepted?

Solution **(i) Present Value of the Lease:**

After Tax Annual Installment = 120,000 (1 – .40) = 72,000

After Tax Interest Rate = 15% (1 – .40) = 9% = .09

$$PV = \frac{A\left\{1 - \frac{1}{(1+R)^N}\right\}}{R}$$

A = Amount of Installment = 72,000
R = Rate of Interest = 0.09
N = Number of Years = 5
PV = Present Value of Installment=?

$$= \frac{72{,}000\left[1 - \frac{1}{(1+.09)^5}\right]}{.09} = \frac{72{,}000 \times .35}{.09} = 280{,}000 \text{ Tk.}$$

(ii) Present Value of Loan:

$$A = \frac{PV \times R}{\left\{1 - \frac{1}{(1+R)^N}\right\}}$$

A = Amount of Installment =?
R = Rate of Interest = 0.15
N = Number of Years = 5
PV = Present Value = 335,000

$$= \frac{335{,}220 \times .15}{1 - \frac{1}{(1+.15)^5}} = \frac{50{,}283}{.50283} = 100{,}000 \text{ Tk.}$$

Loan Payment Schedule

Years	Installment	Beginning balance	Interest	Principal	Ending Balance
1	2	3	4 = (3 × 0.15)	5 = (2 – 4)	6 = (3 – 5)
1	100,000	335,220	50,283	49,717	2,85,503
2	100,000	285,503	42,825	57,175	2,28,328
3	100,000	228,328	34,249	65,751	1,62,577
4	100,000	162,577	24,387	75,613	86,964
5	100,000	86,964	* 13,036	86,964	00

* Rounding Error.

Determination of Present Value of after Tax Net Cash Outflow

Years	Installment	Interest	Depreciation	Interest and Depreciation	Tax (40%)	After Tax Cash Outflow
1	2	3	4	5 = (3 + 4)	6	7 = (2 – 6)
1	100,000	50,283	67,044	117,327	46,931	53,069
2	100,000	42,825	67,044	109,869	43,948	56,052
3	100,000	34,249	67,044	101,293	40,517	59,483
4	100,000	24387	67,044	91,431	36,572	63,428
5	100,000	13,036	67,044	80,080	32,032	67,968

$$\text{Depreciation per year} = \frac{335220}{5 \text{ years}} = 67{,}044 \text{ Tk.}$$

$$PV = \frac{53{,}069}{(1+.09)^1} + \frac{56{,}052}{(1+.09)^2} + \frac{59{,}483}{(1+.09)^3} + \frac{63{,}428}{(1+.09)^4} + \frac{67{,}968}{(1+.09)^5}$$

$$= \frac{53{,}069}{1.09} + \frac{56{,}052}{1.1881} + \frac{59{,}483}{1.2950} + \frac{63{,}428}{1.4116} + \frac{67{,}968}{1.5386}$$

$$= 48{,}687 + 47{,}178 + 45{,}933 + 44{,}933 + 44{,}175$$

$$= 230{,}906 \text{ Tk.}$$

Answer: Present Value of Cash Outflows under:

(i) Lease Financing = 2,80,000 Tk.

(ii) Loan Financing = 2,30,906 Tk.

Decision: Since the present value of outflows under loan financing is less than lease financing, so the company should accept loan financing.